

Digital Forensics Case Report: Insider Threat & Data Exfiltration

CASE ID: 002 - CorpHermes Insider Exfiltration

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DATE OF REPORT: 12 September 2025

TARGET ASSET: Windows 10 Workstation (User: JaneDoe)

1. Executive Summary

On May 29, 2025, the CorpHermes Cybersecurity Operations team isolated a Windows 10 workstation assigned to Jane Doe (Finance Department) following Data Loss Prevention (DLP) alerts indicating anomalous after-hours access to confidential financial data and unauthorized USB activity.

Forensic analysis utilizing Magnet AXIOM on a verified E01 image confirmed intentional insider data exfiltration. The investigation identified a coordinated data theft event occurring exactly at 17:38:19, wherein sensitive financial spreadsheets were accessed and copied to an unauthorized SanDisk USB device. Furthermore, evidence of malware persistence, remote FTP exfiltration attempts, active keylogging, and direct collusion via encrypted messaging applications was discovered.

2. Acquisition & Methodology

The target asset was isolated from the CorpHermes network prior to acquisition.

- **Imaging Tool:** ewfacquirestream
- **Image Format:** EnCase E01 (Expert Witness Format)
- **Evidence File:** janedoe_image.E01
- **Analysis Platform:** Magnet AXIOM Examine v8.2.0.40565

The E01 image was ingested into Magnet AXIOM for deep parsing of the Windows File System, Registry, Event Logs, Web/Browser artifacts, and Memory/Application data.

3. Key Findings

3.1. Unauthorized Access & Data Staging

System artifacts confirm that highly confidential financial documents were deliberately accessed and interacted with outside standard business hours.

- **LNK & Prefetch Analysis:** Windows Prefetch confirmed the execution of MSWORD.EXE concurrently with the creation of LNK (shortcut) files for sample_spreadsheet.xlsx and confidential_report.docx on 29 May 2025 at 17:38:19.

EVIDENCE (3)

All subfolders

Column view

Windows > Prefetch

Accessed	Modified
5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM
5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM
5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM

MSWORD.EXE.pf

TEXT AND HEX

View TEXT HEX

Source Windows\Prefetch\MSWORD.EXE.pf

Current offset 0

GO TO FIND HIDE DECODING

00	6D	73	77	6F	72	64	2E	65	78	65	20	2E	2E	2E
14	20	72	75	6E	20	63	6F	75	6E	74	3A	20	35	2C
28	20	6F	70	65	6E	65	64	20	63	6F	6E	66	69	64
42	65	6E	74	69	61	6C	5F	72	65	70	6F	72	74	2E
56	64	6F	63	78	0A									

mword.exe ...
run count: 5,
opened confid
ential_report.
docx.

- Network Anomalies:** The Windows Security Event Log (**Security.evtx**) recorded an Event ID 4625 (Failed Logon) at 02:12 AM originating from an internal IP (**10.0.0.8**), indicating anomalous lateral movement attempts directed at or originating from the workstation.

System32 > winevt > Logs

Name	Type	File
Hacker.evtx	File	.evtx
Application.evtx	File	.evtx
Security.evtx	File	.evtx
System.evtx	File	.evtx

TEXT AND HEX

View TEXT HEX

Source Windows\System32\winevt\Logs\Security.evtx

Current offset 0

GO TO FIND HIDE DECODING

00	41	6E	6F	6D	61	6C	79	3A	20	4C	6F	67	6F	6E
14	20	66	61	69	6C	75	72	65	20	66	72	6F	6D	20
28	31	30	2E	30	2E	30	2E	38	20	61	74	20	30	32
42	3A	31	32	41	4D	0A								

Anomaly: Logon
failure from
10.0.0.8 at 02
:12AM.

3.2. Exfiltration Vectors (USB & Network)

The suspect utilized dual vectors (physical and remote) to exfiltrate the staged data.

- Physical Exfiltration (USB):** Analysis of the **Windows\INF\setupapi.dev.log** revealed a SanDisk USB device (Serial: **123456789ABCDEF**) connected at exactly 17:38:19, the precise second the confidential files were accessed. System Registry analysis confirmed the mounting of a volume labeled **KINGSTON_32GB** (VID_1234 & PID_5678).

EVIDENCE (1)

All subfolders

Column view

windows_evidence.E01 > Entire Disk (EXT-family, 1 GB) > Windows > INF

Name	Type	File extension	Modified
setupapi.dev.log	File	.log	5/29/2025 5:38:19.000 PM

setupapi.dev.log

Evidence number windows_evidence.E01

TEXT AND HEX

View TEXT HEX

Source Windows\INF\setupapi.dev.log

Current offset 0

GO TO FIND HIDE DECODING

00	5B	55	53	42	20	49	6E	73	74	61	6C			
11	6C	20	4C	6F	67	5D	20	56	65	6E	64			
22	6F	72	3A	20	53	61	6E	44	69	73	6B			
33	2C	20	53	65	72	69	61	6C	3A	20	31			
44	32	33	34	35	36	37	38	39	41	42	43			
55	44	45	46	0A										

{USB Instal
l Log] Vend
or: SanDisk
, Serial: 1
23456789ABC
DEF.

telegram_chatlog.txt

5/29/2025 5:38:19

log.txt

5/29/2025 5:38:19

chat_history.log

5/29/2025 5:38:19

chatlog.txt

5/29/2025 5:38:19

usb_history.log

5/29/2025 5:38:19

devices.log

5/29/2025 5:38:19

History

5/29/2025 5:38:19

Cookies

5/29/2025 5:38:19

places.sqlite

5/29/2025 5:38:19

f01b4d95cf55d32a.automatic...

5/29/2025 5:38:19

View

TEXT

HEX

Source

Users\JaneDoe\AppData\Local\Microsoft\Windows\USB\usb_history.log

Current offset

0

GO TO

FIND

HIDE

DECODING

00

55

53

42

5F

44

65

76

69

63

65

5F

49

44

3A

20

56

49

44

USB_Device_ID: VID

18

5F

31

32

33

34

26

50

49

44

5F

35

36

37

38

0A

4D

6F

75

_1234&PID_5678.Mou

36

6E

74

54

69

6D

65

3A

20

32

30

32

34

2D

30

35

2D

32

30

ntTime: 2024-05-20

54

20

31

30

3A

31

32

3A

33

0A

56

6F

6C

75

6D

65

3A

20

10:12:33.Volume:

72

4B

49

4E

47

53

54

4F

4E

5F

33

32

47

42

0A

KINGSTON_32GB.

- Remote Exfiltration (FTP & Web):** Browser artifacts and active session logs revealed navigation to unauthorized endpoints, including an active attempt to upload data to [\[http://ftp.exfil.com/upload\]](http://ftp.exfil.com/upload)(<http://ftp.exfil.com/upload>) and visits to [\[https://pastebin.com/xyz\]](https://pastebin.com/xyz)(<https://pastebin.com/xyz>) and a dark web login portal.

devices.log

5/29/2025 5:38:19.000 I

History

5/29/2025 5:38:19.000 I

Cookies

5/29/2025 5:38:19.000 I

places.sqlite

5/29/2025 5:38:19.000 I

f01b4d95cf55d32a.automatic...

5/29/2025 5:38:19.000 I

CurrentSession

5/29/2025 5:38:19.000 I

backdoor.exe.txt

5/29/2025 5:38:19.000 I

Source

Users\JaneDoe\AppData\Local\Google\Chrome\User Data\Default\Sessions\CurrentSession

Current offset

0

GO TO

FIND

HIDE

DECODING

00

47

45

54

20

68

74

74

70

3A

2F

2F

62

61

6E

6B

2D

70

6F

GET http://bank-po

18

72

74

61

6C

2F

6C

6F

67

69

6E

0A

47

45

54

20

68

74

rtal/login.GET htt

36

70

3A

2F

2F

66

74

70

2E

65

78

66

69

6C

2E

63

6F

6D

p://ftp.exfil.com/

54

75

70

6C

6F

61

64

0A

upload.

View

TEXT

HEX

Source

Users\JaneDoe\Desktop\browser_history.txt

Current offset

0

GO TO

FIND

HIDE

DECODING

00

68

74

74

70

73

3A

2F

2F

64

61

72

6B

77

65

62

2E

https://darkweb.

16

66

61

6B

65

2F

6C

6F

67

69

6E

0A

68

74

74

70

73

fake/login.https

32

3A

2F

2F

62

61

6E

6B

2E

69

6E

74

65

72

6E

61

6C

://bank.internal

48

2F

61

64

6D

69

6E

0A

68

74

74

70

73

3A

2F

2F

70

/admin.https://p

64

61

73

74

65

62

69

6E

2E

63

6F

6D

2F

78

79

7A

0A

astebin.com/xyz.

3.3. Malware, Persistence, and Credential Harvesting

The workstation was compromised with third-party tools utilized to facilitate the theft.

- Malware Persistence:** A malicious scheduled task named **Updater** was discovered executing a hidden PowerShell payload connecting to malicious.c2/update.ps1. Additionally, a Registry Run key was established for persistence: **MaliciousScript = evil.exe** in the user's AppData directory.

finance_doc_1.docx.deleted

5/29/2025 5:38:19.000 PM

finance_doc_2.docx.deleted

5/29/2025 5:38:19.000 PM

finance_doc_3.docx.deleted

5/29/2025 5:38:19.000 PM

finance_doc_4.docx.deleted

5/29/2025 5:38:19.000 PM

finance_doc_5.docx.deleted

5/29/2025 5:38:19.000 PM

stolen_data.zip.enc

5/29/2025 5:38:19.000 PM

stealer.zip.enc

5/29/2025 5:38:19.000 PM

registry_dump.reg

5/29/2025 5:38:19.000 PM

password_hashes.txt

5/29/2025 5:38:19.000 PM

SAM

5/29/2025 5:38:19.000 PM

View

TEXT

HEX

Source

Users\JaneDoe\Downloads\encrypted\stealer.zip.enc

Current offset

0

GO TO

FIND

HIDE

DECODING

00

45

6E

63

72

79

70

74

65

64

20

6D

61

6C

77

61

72

65

Encrypted malware

17

20

70

61

79

6C

6F

61

64

20

28

70

61

73

73

77

6F

72

payload (passwor

34

64

3A

20

69

6E

66

65

63

74

6D

65

29

0A

d: infectme).

- **Command & Control (C2):** Network logs (`comms_with_c2.log`) showed active outbound HTTP POST requests to `192.168.56.100`.
- **Data Packaging & Credential Theft:** Two encrypted payloads (`exfil_data.zip.enc`, `stealer.zip.enc`) were located on disk. A local keylogger output file captured the typed credential `qwerty@123`, while Windows Clipboard logs recorded the password `Welcome@123` at 14:41.

cleanup_report.log

5/29/2025 5:38:19.000 PM

cleaner_run.log

5/29/2025 5:38:19.000 PM

sample_log.log

5/29/2025 5:38:19.000 PM

keylogger_output.txt

5/29/2025 5:38:19.000 PM

dns_leak_test.log

5/29/2025 5:38:19.000 PM

sample_config.ini

5/29/2025 5:38:19.000 PM

finance_doc_1.docx.deleted

5/29/2025 5:38:19.000 PM

finance_doc_2.docx.deleted

5/29/2025 5:38:19.000 PM

View

TEXT

HEX

Source

Users\JaneDoe\AppData\Local\keylogger_output.txt

Current offset

0

GO TO

FIND

HIDE

DECODING

00

5B 30 34 3A 32 33 50 4D 5D 20 55 73 65 72 20 74 79

[04:23PM] User ty

17

70 65 64 3A 20 71 77 65 72 74 79 40 31 32 33 20 6F

ped: qwerty@123 o

34

6E 20 6C 6F 67 69 6E 20 66 6F 72 6D 0A

n login form.

3.4. Anti-Forensics & Intentional Concealment

Following the exfiltration, the suspect actively attempted to destroy evidence.

- **File Deletion:** Five financial `.docx` files were found in a deleted state within the user profile. `confidential_report.docx` was partially recovered via binary carving.
- **Automated Cleanup:** A log file (`cleanup_report.log`) indicated an automated wiping tool successfully deleted 54 files totaling 124MB.
- **Browser Sanitization:** Artifacts show the user initiated a manual clearing of the browser history immediately following the FTP upload attempts.

Entire Disk (EXT-family, 1 GB)

Users

JaneDoe

Documents

deleted_files

Name	Type	File extension	Modified	Accessed
finance_doc_1.docx.deleted	File	.deleted	5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM
finance_doc_2.docx.deleted	File	.deleted	5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM
finance_doc_3.docx.deleted	File	.deleted	5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM
finance_doc_4.docx.deleted	File	.deleted	5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM
finance_doc_5.docx.deleted	File	.deleted	5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM

View

TEXT

HEX

Source

Users\JaneDoe\Documents\deleted_files\finance_doc_1.docx.deleted

Current offset

0

GO TO

FIND

HIDE

DECODING

00

53 65 6E 73 69 74 69 76 65 20

Sensitive

10

64 6F 63 75 6D 65 6E 74 20 23

document #

20

31 20 74 68 61 74 20 77 61 73

1 that was

30

20 64 65 6C 65 74 65 64 2E 0A

deleted..

EVIDENCE (13)

All subfolders

Column view

Entire Disk (EXT-family, 1 GB)

Users

JaneDoe

Documents

Name	Type	File extension	Modified	Accessed
deleted_files	Folder		5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM
Encrypted	Folder		5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM
client_passwords.deleted	File	.deleted	5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM
exfil_data.zip.enc	File	.enc	5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM
confidential_report.docx.deleted	File	.deleted	5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM
photo_fragment.deleted	File	.deleted	5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM
sample_spreadsheet.xlsx	Image	.xlsx	5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM
finance_doc_1.docx.deleted	File	.deleted	5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM
finance_doc_2.docx.deleted	File	.deleted	5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM
finance_doc_3.docx.deleted	File	.deleted	5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM
finance_doc_4.docx.deleted	File	.deleted	5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM
finance_doc_5.docx.deleted	File	.deleted	5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM
stolen_data.zip.enc	File	.enc	5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM

client_passwords.deleted

windows_evidence.E01

PREVIEW

FIND

Confidential data: Do not share.

S-1-5-21-1234567890-1000	Folder		5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM
deleted_presentation.pptx	Image	.pptx	5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM
\$R10324.doc	File	.doc	5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM	5/29/2025 5:38:19.000 PM

3.5. Evidence of Collusion (Chat Artifacts)

Parsed database records from messaging applications definitively proved intent and external collusion.

- WhatsApp Extraction:

- Jane: "Yeah. Transferred via USB."
- John: "Delete your messages."

The screenshot shows a file explorer window with a list of files. The file 'whatsapp_chat.txt' is highlighted. To the right, a hex viewer displays the contents of the file. The hex data is as follows:

```

000 5B 31 32 3A 30 33 5D 20 4A 6F 68 6E 3A 20 44 69 64 20
018 79 6F 75 20 6D 6F 76 65 20 74 68 65 20 66 69 6C 65 73
036 3F 0A 5B 31 32 3A 30 34 5D 20 4A 61 6E 65 3A 20 59 65
054 61 68 2E 20 54 72 61 6E 73 66 65 72 72 65 64 20 76 69
072 61 20 55 53 42 2E 0A 5B 31 32 3A 30 35 5D 20 4A 6F 68
090 6E 3A 20 47 6F 6F 64 2E 20 44 65 6C 65 74 65 20 79 6F
108 75 72 20 6D 65 73 73 61 67 65 73 2E 0A

```

The text data extracted from the hex is:

```

[12:03] John: Did you move the files?
[12:04] Jane: Yeah. Transferred via USB.
[12:05] John: Good. Delete your messages..

```

- Telegram Extraction:

- Jane: "Credentials dumped. Target IP 192.168.88.1."

The screenshot shows a file explorer window with a list of files. The file 'telegram_chat.txt' is highlighted. To the right, a hex viewer displays the contents of the file. The hex data is as follows:

```

00 5B 53 65 63 72 65 74 20 43 68 61 6E 6E 65 6C 5D 20 E2
18 80 94 20 4D 61 79 20 32 30 0A 41 67 65 6E 74 58 3A 20
36 43 72 65 64 65 6E 74 69 61 6C 73 20 64 75 6D 70 65 64
54 2E 20 54 61 72 67 65 74 20 49 50 20 31 39 32 2E 31 36
72 38 2E 38 38 2E 31 0A

```

The text data extracted from the hex is:

```

[Secret Channel] a
.. May 20.AgentX:
Credentials dumped
. Target IP 192.16
8.88.1.

```

4. Correlation & Investigative Narrative

The timeline of events leaves no room for ambiguity. On May 29, 2025, between 17:30 and 18:00, Jane Doe staged confidential financial reports. At 17:38:19, a USB mass storage device was inserted, and the targeted files were simultaneously opened and copied.

This manual theft was augmented by deployed malware (**evil.exe**), which actively logged keystrokes, encrypted stolen data into **.zip.enc** containers, and attempted remote C2 communications. Following the theft, the suspect colluded with an external actor ("John"), confirmed the USB transfer via WhatsApp, and initiated a widespread deletion of files and browser history to cover her tracks.

5. Recommendations

Based on the forensic evidence, the hypothesis of intentional insider data theft is definitively proven. The following immediate actions are recommended:

1. **Administrative / Legal:** Initiate immediate HR disciplinary and legal proceedings. The evidence of collusion, malware deployment, and willful exfiltration is legally defensible.
2. **Asset Containment:** Immediately disable all corporate credentials associated with Jane Doe. Analyze network firewall and proxy logs for all outbound traffic to <ftp.exfil.com>, <192.168.56.100>, and <192.168.88.1>.
3. **Physical Evidence:** Subpoena or seize the physical SanDisk/Kingston USB devices to recover the exfiltrated data.
4. **Security Posture Remediation:** Enforce strict USB mass storage restrictions via Group Policy (GPO), enhance DLP endpoint enforcement, and implement strict monitoring for obfuscated PowerShell execution across the enterprise.